National University of Computer and Emerging Sciences

Karachi Campus

Artificial Intelligence (AI-2002) Mid I Exam

Date: Wednesday, Sep, 23th, 2026 **Total Marks: 30**

Course Instructor(s) **Total Time (Hrs): 01**

Ms.Alina Arshad, Ms.Saeeda Kanwal **Total Questions: 03**

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Roll No **Section** **Student Signature**

Do not write below this line. Attempt all the questions in-order.

CLO # 1,2,3

Q.1 : [marks: 10] **[15 minutes]**

Texas Hold'em Poker is a card game played by 2–6 players. Each player is dealt two private ("hole") cards, unseen by the other players. Five shared community cards are revealed during the game: the flop (three cards), turn (one card), and river (one card). Players may fold, check, call, or raise during the betting rounds. At showdown, the remaining players reveal their hole cards, and the player with the best five-card hand wins the pot.

a. Classify the poker-playing environment according to each of the following properties: Fully observable/Partially observable, Deterministic/Stochastic, Episodic/ Sequential, Static / Dynamic, Discrete/Continuous, Single-agent/Multi-agent. For every classification, provide a specific justification based on the mechanics of Texas Hold'em. A bare label without justification will receive no marks. [6]

b. Give the PEAS description for a poker-playing intelligent agent.[4]

Q.2 : [marks: 10] **[20 minutes]**

A merchant caravan must travel from Samarkand (A) to Kashgar (H) along the trade-post network shown in Figure. Edge weights are toll costs in trade units not distances; do not infer costs from map geometry. A straight-line estimate h(n) to Kashgar is given below.

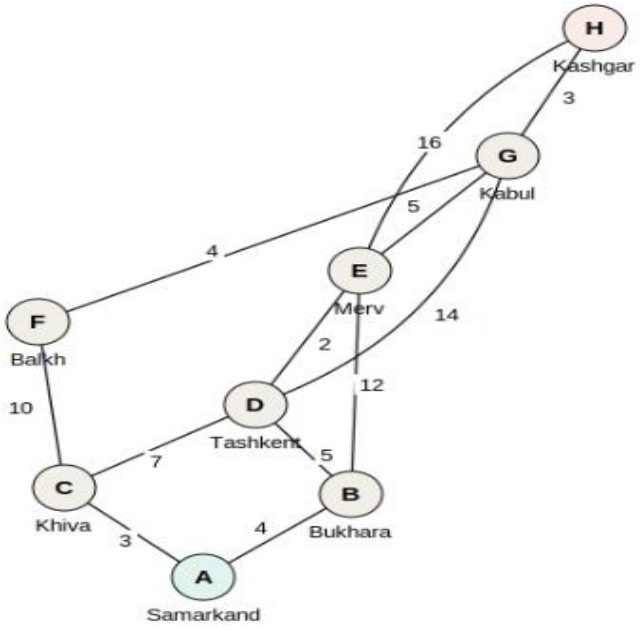
n	A	B	C	D	E	F	G	H
h(n)	14	12	11	9	7	6	3	0

Assume graph search (no repeated states), neighbours generated in alphabetical order, ties in path-cost/g broken alphabetically, and for A*, ties in f broken in favour of the smaller h (then alphabetically).

a. Run BFS. Give the expansion order, the path returned, and its cost. State precisely what quantity BFS is guaranteed to minimize, and why that guarantee does not make this path optimal.

b. Run DFS. Give the expansion order, the path returned, and its cost.

c. Run Uniform Cost Search. Show the frontier (node, g) after every expansion, marking every g-value update. Give the expansion order and the optimal path with cost
d. Run A*. Give a full trace with g, h, f at each expansion. State exactly which node UCS expands that A* does not, and explain what property of h causes this saving.
e. Someone proposes h(D) = 13 instead of 9, arguing it makes the search "more confident" about avoiding D. Re-run A* with only this change, give the resulting path and cost, and name precisely which theoretical guarantee is now broken.



Q.3: [marks: 10] **[20 minutes]**

A smart farming system has 6 agricultural fields, each with a different initial productivity value. Three types of crops: Wheat, Rice, and Vegetables have to be assigned to different fields.

Field	Productivity Value
F1	30
F2	25
F3	18
F4	12
F5	8
F6	5

Wheat is highly suitable for a field, so whichever field is assigned Wheat, its productivity becomes 1.5 times its initial value; if the result is decimal, take the floor value. Rice has a normal effect, so the productivity of the field assigned to Rice remains **unchanged**. Vegetables are less suitable, so whichever field is assigned Vegetables, its productivity becomes half of its initial value, taking the floor if the result is decimal.

Your task is to simulate a dry run of genetic algorithms in this case running for 2 iterations.Choose an initial population of four. Next apply a fitness function to your problem. Choose the best three subsequently. Apply single-point crossover point and then apply mutation. For mutation we have been told that chances of mutations are: 200 mutations in 100 generations.

Solution:

Q.1

a. Partially observable: hidden hole cards and unrevealed deck.
Stochastic: random card dealing.
Sequential: previous actions affect future states, pot size, and decisions.
Dynamic : other players can change the game state through their actions.
Discrete : cards, actions, betting decisions, and game states are discrete/countable.
Multi-agent : multiple independent decision-making players.

b. **Performance measure:** Net chips won over a session (or tournament survival/placement); may also weight variance reduction for a risk-averse agent.
Environment: The table: opponents, the deck, community cards, pot, betting structure, blinds.
Actuators: Fold, call, check, raise (with an amount), and any table-interface actions (e.g. showing/mucking a hand).
SensorsOwn hole cards, revealed community cards, visible chip stacks, opponents' bets and timing, and (in physical poker) opponent behaviour/"tells"

Marking scheme: 1 each

Q.2

a. Expansion order: A → B → C → D → E
Path: A → B → E → H
Cost: 4 + 12 + 16 = 32, BFS finds a solution with the fewest number of edges, not necessarily the lowest cost. In this case, A → B → E → H has only 3 edges, and there is no path from A to H with fewer edges, so BFS correctly selects this path. However, because the edge costs are different (ranging from 2 to 16), the shortest path in terms of edges is not necessarily the cheapest path. The actual optimal path has a cost of 19, while the BFS path costs 32, which is about 68% more expensive. So, BFS did exactly what it is designed to do: it minimized the number of steps, not the total path cost.

0.5 each part

b. Expansion order: A, B, D, C, F, G, E, H
Path: A → B → D → C → F → G → E → H, cost = 4+5+7+10+4+5+16 = 51, DFS gives no completeness or optimality guarantee of any kind on this graph — it simply returns the first path found by alphabetical depth-first exploration.

0.5 each part.

c. Expansion order: A, C, B, D, E, F, G, H
Optimal path: A → B → D → E → G → H, cost = 4+5+2+5+3 = 19

0.5 for expansion order and optimal path, 1 for cost

d. Expansion order: A, C, B, D, E, G, H — 7 nodes. Path: A → B → D → E → G → H, cost 19 , F is the node UCS expands that A* does not. UCS expanded F because 13 was simply the cheapest g on its frontier at that point; A* deferred it because h(F)=6 correctly signals F is still far from Kashgar, so F's f=19 never becomes competitive before the true solution (also f=19) is found. This saving is possible because h is consistent: f never decreases along any path, so the first time a node is selected it already has its optimal g, and any node with f exceeding the eventual solution cost can be safely skipped.

e. (D reaches g=9 via B but f=22, so it's never competitive; D and E are both starved out.)Path returned: A → C → F → G → H, cost 20 — one unit worse than the true optimum of 19.True cost from D to Kashgar is 10, so h(D)=13 overestimates it: the heuristic is no longer admissible (and, checking |13−7|=6 > c(D,E)=2, no longer consistent either). A*'s optimality guarantee requires h ≤ h*; making an estimate larger does not make it "safer," it removes the guarantee outright, and here that removal produces a genuinely wrong answer rather than just a theoretical risk.

Q.3 Given Data have different valid initial populations, crossover points, and mutation positions can produce different intermediate chromosomes. The important requirement is that the GA procedure is applied correctly and consistently.

Field	Initial Productivity
F1	30
F2	25
F3	18
F4	12
F5	8
F6	5

The effects are:

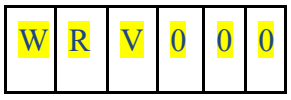
- Wheat: productivity × 1.5
- Rice: productivity remains unchanged
- Vegetables: productivity ÷ 2, taking floor

Field	Wheat	Rice	Vegetables
F1	45	30	15
F2	37	25	12
F3	27	18	9
F4	18	12	6
F5	12	8	4
F6	7	5	2

Wheat → F1 = 45, Rice → F2 = 25, Vegetables → F3 = 9
Therefore:
Fitness = 45 + 25 + 9 = 79

Step 1: Chromosome Representation
We represent each solution as:
[Wheat, Rice, Vegetables]
For example:
[F2, F3, F4]
means:

- Wheat → F2
- Rice → F3
- Vegetables → F4



OR MOST SUITABLE

F1 F2 F3 F4 F5 F6

Step 2: Initial Population
Assume the following initial population of four chromosomes:

Chromosom	Representatio n

C1	[F1, F2, F3]
C2	[F2, F3, F4]
C3	[F3, F1, F5]
C4	[F4, F5, F6]

Step 3: Calculate Fitness

Chromosome C1

C1 = [F1, F2, F3]

Wheat → F1:

30 × 1.5 = 45

Rice → F2:

25

Vegetables → F3:

floor(18 ÷ 2) = 9

Therefore:

Fitness(C1) = 45 + 25 + 9 = 79

Chromosome C2

C2 = [F2, F3, F4]

Wheat → F2:

floor(25 × 1.5) = 37

Rice → F3:

18

Vegetables → F4:

floor(12 ÷ 2) = 6

Therefore:

Fitness(C2) = 37 + 18 + 6 = 61

Chromosome C3

C3 = [F3, F1, F5]

Wheat → F3:

18 × 1.5 = 27

Rice → F1:

30

Vegetables → F5:

floor(8 ÷ 2) = 4

Therefore:

Fitness(C3) = 27 + 30 + 4 = 61

Chromosome C4
C4 = [F4, F5, F6]
Wheat → F4:
12 × 1.5 = 18
Rice → F5:
8
Vegetables → F6:
floor(5 ÷ 2) = 2
Therefore:
Fitness(C4) = 18 + 8 + 2 = 28

Initial Population Fitness

Chromosome	Representation	Fitness
C1	[F1, F2, F3]	79
C2	[F2, F3, F4]	61
C3	[F3, F1, F5]	61
C4	[F4, F5, F6]	28

Step 4: Selection
We have to select the best three chromosomes.
The ranking is:

1. C1 = 79
2. C2 = 61
3. C3 = 61
4. C4 = 28

Therefore, selected chromosomes are:
C1, C2, C3
C4 is eliminated.

Step 5: Crossover
We need to select a crossover point.
Let us choose:
Crossover point = after the first gene
So:

C1 = [F1 | F2 F3]
C2 = [F2 | F3 F4]
Exchange the portions after the crossover point.
Offspring 1
From C1 take the first gene:
F1
From C2 take the remaining genes:
F3 F4
Therefore:
O1 = [F1, F3, F4]
This is valid because all three fields are different.
Offspring 2
From C2 take:
F2
From C1 take:
F2 F3
This produces:
[F2, F2, F3]
This is invalid because F2 appears twice.
We therefore repair the duplicate by replacing the repeated field with a missing field.
The fields already present are F2 and F3. The missing field can be F1.
Therefore:
O2 = [F2, F1, F3]

Step 6: Fitness of Offspring
O1 = [F1, F3, F4]
Wheat → F1:
45
Rice → F3:
18
Vegetables → F4:
6
Therefore:
Fitness(O1) = 45 + 18 + 6 = 69

O2 = [F2, F1, F3]
Wheat → F2:
37

Rice → F1:
30
Vegetables → F3:
9
Therefore:
Fitness(O2) = 37 + 30 + 9 = 76

Step 7: Mutation
The question states:
200 mutations in 100 generations
Therefore:
200/100=2
So, we have:2 mutation operations per generation.
For this dry run, we will perform exactly two mutation operations in each generation.
For Generation 1, suppose the following two mutations are selected.

Mutation 1
Take O1:
O1 = [F1, F3, F4]
Mutate the third gene:
F4 → F5
New chromosome:
O1' = [F1, F3, F5]
Its fitness is:

- Wheat → F1 = 45
- Rice → F3 = 18
- Vegetables → F5 = 4

Therefore:
Fitness = 45 + 18 + 4 = 67

Mutation 2
Take O2:
O2 = [F2, F1, F3]
Mutate the second gene:
F1 → F4
New chromosome:
O2' = [F2, F4, F3]

Its fitness is:

- Wheat → F2 = 37
- Rice → F4 = 12
- Vegetables → F3 = 9

Therefore:
Fitness = 37 + 12 + 9 = 58

Generation 2 Population
For the next generation, we retain the best parent as an elite and add the offspring.
Let the Generation 2 population be:

Chromosome	Representation	Fitness
C1	[F1, F2, F3]	79
O1'	[F1, F3, F5]	67
C3	[F3, F1, F5]	61
O2'	[F2, F4, F3]	58

Generation 2: Selection
Rank the chromosomes:

1. C1 = 79
2. O1' = 67
3. C3 = 61
4. O2' = 58

Therefore, the best three are:
C1, O1', C3

Generation 2: Crossover
We again use a crossover point after the first gene.
Take:
C1 = [F1 | F2 F3]
and
O1' = [F1 | F3 F5]
After crossover:
Offspring 3
First part from C1:
F1

Remaining part from O1':
F3 F5
Therefore:
O3 = [F1, F3, F5]
This is valid.
Fitness:

- Wheat → F1 = 45
- Rice → F3 = 18
- Vegetables → F5 = 4

Fitness(O3) = 45 + 18 + 4 = 67
Offspring 4
First part from O1':
F1
Remaining part from C1:
F2 F3
Therefore:
O4 = [F1, F2, F3]
Fitness:
45 + 25 + 9 = 79

Generation 2: Mutation
Again, according to the given mutation information, we perform 2 mutation operations.
Mutation 1
Take O3:
O3 = [F1, F3, F5]
Change the third field:
F5 → F6
New chromosome:
O3' = [F1, F3, F6]
Fitness:

- Wheat → F1 = 45
- Rice → F3 = 18
- Vegetables → F6 = 2

Therefore:
Fitness = 45 + 18 + 2 = 65

Mutation 2
Take O4:
O4 = [F1, F2, F3]
Change the second field:
F2 → F4
New chromosome:
O4' = [F1, F4, F3]
Fitness:

- Wheat → F1 = 45
- Rice → F4 = 12
- Vegetables → F3 = 9

Therefore:
Fitness = 45 + 12 + 9 = 66

Final Population After 2 Iterations
The resulting population can be shown as:

Chromosome	Wheat	Rice	Vegetables	Fitness
C1	F1	F2	F3	79
O1'	F1	F3	F5	67
C3	F3	F1	F5	61
O4'	F1	F4	F3	66

The highest fitness is:79
Therefore, the best chromosome is:
[F1, F2, F3]
which means:

- Wheat → F1
- Rice → F2
- Vegetables → F3

Its total fitness is:
45+25+9=7945+25+9=79
Final Answer
Best assignment after two iterations:

Crop	Assigned Field	Contribution

Wheat	F1	45
Rice	F2	25
Vegetables	F3	9
Total Fitness		79